Variables of the Coded PTA 6–10 Files

31 October 2026 · August Liu. The coding rules are in the workflow document; results are in the result report.

Files

File	Agreement	Rows
TUR_SGP_coded_2026_10_31.dta	Turkey–Singapore	21,873
NIC_TWN_coded_2026_10_31.dta	Nicaragua–Taiwan	211,027
CHL_KOR_coded_2026_10_31.dta	Chile–Korea	313,785
OMN_USA_coded_2026_10_31.dta	Oman–US	249,838
CAN_CHL_coded_2026_10_31.dta	Canada–Chile	374,528
Total		**1,171,051**

One row is one (restricted input, output) pair from the agreement's product-specific rules of origin, in the order of the original files. Turkey–Singapore is in HS2012; the other four are in HS2002. Every pair is coded on HS2002 (input_hs2002, output_hs2002). The files share 36 variables, each with a Stata label; Turkey–Singapore has one more, special_rule, from its original file. They can be combined with `append`, which widens string variables as needed:

```
use TUR_SGP_coded_2026_10_31, clear
foreach p in NIC_TWN CHL_KOR OMN_USA CAN_CHL {
    append using `p'_coded_2026_10_31
}
```

A pair (master_id) has the same DV in every agreement where it appears, and the same DV as in the PTA 1–5 files. 3,004 rows map to several HS2002 pairs. The 2,591 coded on HS2002 take the highest DV among their pairs; the other 413 are Turkey–Singapore rows coded directly on their HS2012 codes. So 2,143 HS2002 pairs with DV = 0 fall inside such a row with DV = 1; 801 of them are also coded 0 on rows of their own.

Variables

Variable	Type	Contents
output_hs6	str6	Output, 6-digit code in the agreement's HS vintage (the original output_str)
output_description	strL	Output: heading text | subheading text, in the agreement's HS vintage
input_hs6	str6	Restricted input, 6-digit code in the agreement's HS vintage (the original input_str)
input_description	strL	Input: heading text | subheading text
VA_Alternative, VA_Complement	byte	From the original agreement file, unchanged
special_rule	byte	Turkey–Singapore only: from the original file, unchanged
pta	str	Agreement: TUR_SGP, NIC_TWN, CHL_KOR, OMN_USA, CAN_CHL
record_no	long	Row number in the original file
pair_id	str	Unique row id: pta|input>output|record_no
hs_vintage	str	HS vintage of input_hs6 and output_hs6 (HS2012 for TUR_SGP, HS2002 for the others)
input_type	str	hs6 for every row (rows whose input was "WO" or "any" were deleted)
input_hs2002, output_hs2002	str	The codes mapped to HS2002; several codes are separated by ";"
n_mapped_pairs	int	Number of HS2002 pairs the row maps to (1 for most rows)
mapped_DV1_share	double	Share of those HS2002 pairs with DV = 1
concord_flag	byte	1 = a code is not one-to-one with HS2002, or not a valid code (5,235 TUR_SGP rows, 614 CHL_KOR rows)
concord_note	strL	Type of the concordance link, e.g. "input HS2012 n to 1", "output invalid code: all HS2002 subheadings of 61034"; empty when one-to-one
DV	byte	**Main variable.** 1 = the input can be used to make the output (including links that are possible but unconfirmed, marked uncertain); 0 = no link
DV_strict	byte	Robustness version: DV × (1 − uncertain), only certain links
DV_excl_remelt	byte	Robustness version: as DV, but direct remelting (route = recycle_direct, e.g. steel rod → steel ingots) = 0, per HS2002 pair; a row coded on HS2002 that maps to several HS2002 pairs takes the highest value among them (so 3 TUR_SGP rows shown as recycle_direct keep 1, because another of their pairs has another route); the 591 TUR_SGP rows coded on their HS2012 codes are coded directly
DV_max	byte	Same as DV (kept for older code)
uncertain	byte	1 = kept at DV = 1, but the link is possible rather than known; the reason is in uncertain_reason
uncertain_reason	strL	Why uncertain = 1 (list below); empty when uncertain = 0
locked	byte	1 = locked at workflow step 2 as the same good (most, not all, rows with route = same_good)
step	double	Workflow step that set DV = 0: 3 unrelated goods, 4 parallel finished goods, 5 reverse direction, 6 no route; 2 = same good, certain (route = same_good, uncertain = 0); missing when DV = 1 through another route, or when a same_good pair is uncertain

route	str	Why the pair is 1, or none (list below)
source	str	How the code was obtained (list below)
copy_from	str	For copies: ASEAN_JPN (ASEAN–Japan labels) or PTA_2_5 (Brunei–Japan, Australia–US, CAFTA, Bahrain–US); empty otherwise
policy	strL	Decisions and conventions that changed the pair, separated by ";" (list below); empty when none did
ml_prob	double	Model probability that DV = 1 (information only); filled for the newly coded HS2002 pairs (source ai_units, ai_sample, rule_textile, rule), missing for copies of PTA 1–5 and for ai_concord
ml_supported	double	1 = the model had enough similar coded pairs to be used (information only); filled for the same rows as ml_prob
dv_ai1, dv_ai2	double	Code of AI coder 1 (Claude) and AI coder 2 (Codex) where they coded the pair; otherwise missing
why_ai1, why_ai2	strL	The two AI coders' one-sentence reasons
master_id	str	Id of the HS2002 input–output pair (M|input>output), the same in every agreement and in the PTA 1–5 files; HS2012|input>output for pairs coded on HS2012 codes. Of the 3,004 rows that map to several HS2002 pairs, the 2,591 coded on HS2002 (1,977 TUR_SGP, 614 CHL_KOR) have as master_id one of those pairs, whose DV equals the row's DV; the other 413 TUR_SGP rows are coded on their HS2012 codes (master_id HS2012|…)

Values of route

Value	Meaning
`same_good`	Same good, or the same good further processed or packed (step 2 when certain; mostly locked)
`forward`	The input is an earlier stage of the output (e.g. yarn → fabric of the same fibre)
`ingredient`	The input goes into the output as an ingredient, constituent or process input
`component`	The input is a part of the output (e.g. nibs → pens, trims → garments)
`mixture`	The input is one component of a mixture or blend
`sibling_intermediate`	Two intermediate goods of the same material
`recycle_waste`	Through a waste or scrap good
`recycle_direct`	Direct remelting or re-processing with no scrap code in between (0 in DV_excl_remelt, per HS2002 pair)
`feed`	The input feeds a farmed animal (e.g. feed fish → farmed salmon)
`reconstitute`	The output is the input reconstituted (e.g. dried → liquid egg albumin)
`ml_forward`, `ml_ingredient`, `ml_component`, `ml_recycle`, `ml_sibling`	Copied textile pairs kept at 1 by the model and confirmed by the textile rules in PTA 2–5; the route predicted by the model
`unresolved`	Open case in the ASEAN–Japan labels (DV = 1, uncertain = 1)
`none`	No link (DV = 0)

Values of source

Value	Meaning
`exact_copy`	Copied from the PTA 1–5 codes (same HS2002 pair); copy_from says from which
`ai_units, ai_sample`	The two AI coders (Claude and Codex): heading-pair units, and a blind sample of textile pairs
`ai_concord`	TUR_SGP rows where two HS2012 codes fall into the same HS2002 code: coded by the two AI coders on the HS2012 codes
`rule_textile`	Explicit textile rules (chapters 50–63)
`rule`	Input and output have the same code

Values of uncertain_reason

Value	Meaning
AI coders, or the model and the textile rules, disagree	One AI coder found a production route and the other did not (or the model and the textile rules disagreed), no convention settles the pair, and the screen did not find the 1 clearly invalid
AI coders disagree: sibling or backward	One coder saw two intermediate siblings, the other a reverse (backward) pair
same-material intermediate of another heading, or another compound (convention of 2026-10-30)	Parts or metal articles of another heading or material, or another compound of the same chemical heading, linked only as same-material intermediates
remelting into another primary or semi-finished form (convention of 2026-10-30)	Remelted or re-processed into another primary or semi-finished form of the same material (not the input's own upstream form)
remelting excluded (convention of 2026-10-30), another route possible	The remelting route is excluded, but a screen named another specific route (component, ingredient, forward)
minor constituent of a garment or made-up: other fibre or construction	Yarn or fabric of a fibre other than cotton or man-made fibres (wool, silk, flax, ramie, coarse hair, other), or of the same fibre but another construction (woven ↔ knitted), in a garment or made-up article
minor constituent of a yarn or fabric: other fibre	Fibre, yarn or waste of a fibre other than cotton or man-made fibres as a minor blend component of a yarn or fabric
open case in the ASEAN-Japan labels	Left open in the original ASEAN–Japan hand coding (route = unresolved); copied
HS concordance: the HS2002 pairs of this record differ	The row maps to several HS2002 pairs whose DV differ; the row takes DV = 1
possible feed (rock lobster)	Crustaceans or crustacean meal → rock lobster: a possible but not established feed use

Values of policy

Conventions of 19 October 2026 (on copied pairs, as in the PTA 1–5 files; P_Q1_1 and P_Q4 also mark newly coded pairs, where the conventions of 30 October then refine them: P_Q2C or P_2CFLAG after P_Q1_1, mostly P_Q3E_* after P_Q4) and of 28–30 October 2026 (on newly coded pairs only); see the summary table of the workflow document.

Value	Convention
P_Q1_1	Intermediate goods of the same material at the same stage = 1 (pairs on which the AI coders disagreed)
P_Q1_2	Different synthetic fibres count as one fibre
P_Q2_1	Any chemical (chapters 28–38) → wastes of heading 3825 = 1
P_Q2_2	Any mineral of chapter 25 → cement (2523) = 1
P_Q4	Direct remelting = 1 in DV (route recycle_direct), before the limits of 30 October
P_CONF, P_UNC	Copied textile pairs where the model and the textile rules disagreed (P_CONF), or kept at 1 and marked uncertain as a minor constituent (P_UNC)
P_AUD_A, P_AUD_B, P_AUD_C	Audit corrections of 19 October: minor fibres in yarn or fabric = 1 with uncertainty; intermediate goods of one material coded 0 by both coders = 1; sewing thread ↔ yarn = 1
P_OPT2	Cotton or man-made fibre, yarn, fabric or waste as a minor constituent = 1 (certain)
P_FISH	Fish and other seafood of different species = 0 (step 3), unless the input could feed the output
P_Q3E_own	Remelting back to the input's own upstream form = 1 (30 October)
P_Q3E_other	Remelting into another primary or semi-finished form of the same material = 1 with uncertainty
P_Q3E_0	Remelting into a finished product, vulcanised rubber, different or non-meltable polymers = 0
P_RESCUE	A P_Q3E_0 pair with another specific route = 1 with uncertainty
P_Q2C, P_Q2C_same	Parts and metal articles of another heading or material, or another compound of one heading = 1 with uncertainty (P_Q2C); metal articles of one heading and metal = 1 (P_Q2C_same)
P_2CFLAG	Uncertain flag only: different compounds of one heading claimed as intermediates, and same-material intermediates that P_Q1_1 had made certain
P_Q5B	Swine meat, offal and fat → pig bristles = 0 (parallel by-product)
P_A1	One-sided 1 that both model families found clearly invalid = 0
P_PROF0	Both independent auditors coded 0, in a class where Professor Zhang said such pairs are 0